Non-monotonic source reversion across language-model pretraining checkpoints: behavioral trajectories and source-position residual interventions

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Abstract

Whether a language model continues to follow explicit evidence during training is often reported primarily at a final checkpoint. We trace candidate choices across checkpoints and test local residual replacement. Conflict prompts state counterfactual `X` while a parametric prior favors `Y`; corrupt controls replace `X` with `Z`. Across 24 OLMo-2 1B checkpoints, the largest observed low-high-low prior-choice reversion excursion is 0.434 (95% CI [0.362, 0.523]); a separate eight-checkpoint 7B release shows 0.230 (95% CI [0.158, 0.298]). At the selected 1B transition, the $p_X, p_Y, p_{\text{other}}$ decomposition changes from (0.390, 0.470, 0.140) to (0.762, 0.182, 0.056); the 7B decomposition is concordant (Table 3). Early-depth source-donor patch effects increase by +1.054 [0.854, 1.255] and +1.008 [0.802, 1.215] single-token X-Y logit-difference units for two of three 1B templates; corresponding 7B changes are +0.968 [0.752, 1.191] and +0.672 [0.469, 0.877]. At prespecified frozen semantic peaks, conditional descriptive summaries from the gated suite are positive across type/category, relation/slot, and naturalistic axes in Amber-7B and Pythia-6.9B. The results identify a bounded source/prior reversion pattern and a causal effect of residual replacement at the tested source position. Nontrivial Pythia random-residual controls leave source-content specificity unresolved; no universal training phase is established.

Keywords: language models; training dynamics; contextual faithfulness; knowledge conflict; activation patching; mechanistic interpretability

1. Introduction

Language models are often asked to answer from an explicit passage while also carrying a parametric prior acquired during pretraining. When the passage and the prior disagree, an output can be fluent and plausible while failing to follow the evidence that was supplied in the prompt. This distinction is central to faithfulness evaluation in generation [1] and to recent studies of contextual knowledge conflict [2,3]. It is also a training question: a model's final behavior need not reveal whether source use was present, lost, or recovered earlier in the optimization trajectory.

Public checkpoint suites make this question measurable. Pythia was released with models trained on the same data order and many intermediate checkpoints [4], while OLMo and OLMo 2 provide open weights, training information, code, and intermediate checkpoints [5,16]. Earlier work on delayed generalization and mechanistic progress measures shows why a single score can hide gradual or non-monotonic changes [6,7]. Recent work now also traces parametric and in-context preference during pretraining, including Pythia checkpoints and an OLMo-7B appendix [17]. We therefore ask a narrower unresolved question: can a controlled explicit-source conflict show a low–high–low reversion excursion within an unmodified OLMo-2 trajectory, and does a selected behavioral change have a local causal activation effect?

Behavior alone is not enough to identify source transmission. A representation may contain information that a model does not use for its output, and a probe can measure availability without establishing behavioral influence [8]. Recent work on latent knowledge and truthfulness likewise reports dissociations between internal encoding and generated answers [9-11]. Causal mediation and activation-patching methods address this distinction by replacing an intermediate activation and measuring the resulting output change [12-15]. The intervention still has to be defined locally: a source-token patch can test the causal effect of replacing the intervened residual on a specified margin, but it does not by itself identify a complete circuit or explain why a training transition occurs.

We study these questions with a matched source-reversion task and a source-position residual intervention. The conflict prompt states a counterfactual answer `X` while the catalogued factual answer `Y` serves as the parametric candidate; its source-erased preference is measured directly. A matched corrupt prompt replaces only the source answer with `Z`. We first trace reversion across OLMo-2 1B and 7B checkpoints. We then patch the clean `X` source-token residual into the corrupt run at each decoder block. Finally, at frozen semantic peaks selected by the independent-family gate, we test type/category, relation/slot, and naturalistic context axes in Pythia-6.9B and Amber-7B, with prespecified deterministic item-key halves and donor controls.

The paper makes three primary contributions. First, it quantifies a reproducible low–high–low trajectory of parametric-prior reversion and verifies that the corresponding source-choice proportion moves in the opposite direction rather than being explained by distractor choices. Second, it measures a causal change in the effect of replacing the answer-bearing source-position residual at selected transitions. Third, it tests the same source-versus-corrupt construction at three semantic axes in two independently trained families at frozen checkpoints. We additionally report the failed pre-final predictor/steering gate rather than converting an exploratory correlation into a control method. The resulting claim is intentionally bounded: it concerns the tested checkpoints, prompts, source positions, and interventions, not a universal phase boundary or a general law of language-model training.

2. Related work

2.1 Faithfulness and contextual knowledge conflict

Faithfulness is commonly defined relative to an input document or source: an answer should be supported by the supplied evidence rather than hallucinated from unrelated model knowledge [1]. In language models, the analogous failure appears when parametric knowledge competes with an explicit context. Zhou et al. show that prompting can improve context use in such conflicts [2], and Xie et al. provide a controlled study of how model behavior varies with the coherence and strength of external evidence [3]. Kim et al. [17] recently traced parametric and in-context preference during pretraining and showed that the two source preferences need not exhaust all outputs. Our source-reversion analysis uses candidate-string scoring to ask whether the source/prior choice balance can move in both directions within an unmodified OLMo-2 trajectory; the causal follow-up then uses a one-token source/corrupt contrast and a local residual intervention.

2.2 Training dynamics and checkpoint-wise analysis

The Pythia release was designed for studies of how language models develop across training and scale [4]. OLMo and OLMo 2 similarly make intermediate model states available for scientific analysis [5,16]. Work on grokking demonstrates that generalization can occur after apparent memorization [6], and Nanda et al. show that mechanistic progress measures can expose continuous changes beneath an abrupt behavioral curve [7]. We use this perspective as a methodological analogy, not as evidence that the present source-reversion curve has the same algorithmic phases as modular arithmetic. In particular, the current data do not justify a universal three-phase description. She et al. [19] likewise show that linear steerability can emerge at intermediate checkpoints and vary across concepts and model families; our failed prospective gate and local patching analysis address a different source-versus-corrupt margin and do not assume that steerability is a reliable predictor.

2.3 Representations, behavior, and causal interventions

Amnesic probing formalized the distinction between information present in a representation and information that contributes to a task decision [8]. Latent-knowledge work similarly asks what a model encodes when its generated answer is wrong [9-11]. Causal mediation, causal tracing, and interchange interventions go further by modifying an internal variable and measuring an output counterfactual [12-14]. Activation-patching best-practice work stresses that corruption choice, metric, and intervention geometry can change the interpretation [15]. Makelov et al. [20] further show that subspace patches can produce an end-to-end effect without uniquely identifying the intended feature. Our design follows these lessons by requiring an exactly one-token clean/corrupt difference, retaining the corrupt donor as a no-op control, and reporting wrong-position and norm-matched random-residual donors separately.

2.4 Position of the present study

The contribution is the combination of three elements: a checkpoint-wise source-reversion trajectory with an explicit X/Y/other decomposition, a source-position intervention at selected training transitions, and a frozen-peak semantic suite with explicit falsification checks. This differs in question and intervention from recent work on training-data factors and source-preference curves [17]. The ingredients are combined to provide a reproducible empirical account of how an explicit source affects a measured answer margin under a controlled conflict construction; source reversion is treated as that candidate-choice construct rather than as a synonym for factuality, truthfulness, or general reasoning.

3. Experimental design

3.1 Source-reversion task and metrics

Each primary item is a factual tuple with a subject, a source-stated answer, and a candidate list. In a conflict prompt, the passage asserts a counterfactual answer `X`, while the model's parametric knowledge is evaluated against the catalogued answer `Y`. The candidate list also contains distractors. A matched source-erased prompt removes the answer-bearing statement and supplies a continuous prior-strength diagnostic. A neutral prompt uses a fictional or otherwise prior-minimized subject without a catalogued factual prior while retaining the same source-answer construction. The pilot item generator also defines an agree condition, but no agree measurements are part of the canonical OLMo trajectory or semantic confirmation runs; it is not used as evidence below.

The information flow is therefore:

```
Parametric prior: Y    Clean source: X    Corrupt source: Z
Candidate set: X, Y, and distractors
```

Only the source answer changes between the clean and corrupt prompts in the causal runs; the candidate set and query remain fixed.

For the primary identity axis, we score the candidate strings with length-normalized autoregressive log likelihood. Let $\ell_c(a)$ and $\ell_e(a)$ denote the candidate score under conflict and source-erased prompts. The reversion indicator is

$$r_i = \mathbf{1}\{\arg\max_a \ell_c(a) = Y_i\},$$

so a higher reversion rate means more frequent selection of the parametric prior. Because distractors can win, a lower reversion rate is not by itself a source-faithfulness rate. We therefore also report

$$p_X = \Pr(\arg\max_a \ell_c(a) = X), \quad p_Y = \Pr(\arg\max_a \ell_c(a) = Y), \quad p_{\text{other}} = 1 - p_X - p_Y.$$

The source-choice proportion p_X is the direct candidate-level readout of source following; p_{other} records distractor choices rather than silently assigning them to either source. We also report the conflict margin $m_c = \ell_c(X) - \ell_c(Y)$, the erased margin $m_e = \ell_e(X) - \ell_e(Y)$, and the source contribution $d_m = m_c - m_e$. A descriptive prior-strength quantity is the erased score of `Y` relative to the distractor mean. These continuous quantities are not treated as causal predictors.

The template-level behavioral audit and the causal batch additionally retain the pairwise Y-over-X error $e_{YX} = \mathbf{1}\{m_c < 0\}$. This indicator compares only `X` and `Y`; it can disagree with `r_i` when a distractor is the top-scoring candidate. We label it explicitly wherever those template or causal-batch values are reported.

Candidate scores are autoregressive log-likelihoods for the continuation `" " + candidate`, with no appended end-of-sequence token, divided by the number of continuation tokens (and by one for an empty continuation). The primary identity trajectory therefore uses normalized log-likelihood differences in nats per continuation token; candidates are not required to be single tokens in this trajectory. The source-position causal batch separately filters `X`, `Y`, and `Z` to one first-answer token. Its final-token logit difference is numerically the corresponding one-step log-odds (nats), but it is not the same estimator as the length-normalized trajectory score. The candidate menu is scored in the same answer slot for the conflict and erased prompts; tokenizer revisions are pinned in the run manifests.

Randomized roster tags are presentation-only. They remain in the prompt context, but are excluded from every scored continuation: the scorer appends only `" " + candidate` (or, in the one-token causal batch, scores the answer token itself). Thus tag identity cannot contribute token likelihood to a candidate score.

For the primary identity bank, the generator samples 40 matched conflict/erased pairs in each of two categories (`loc`, `auth`) and two fame strata (160 rows per checkpoint). The factual banks contain 16 location and 12 authorship facts; repeated samples collapse to 121 unique (`category`, `subject`, `X`, `Y`) clusters for statistical inference and bootstrap resampling. In a conflict pair, `Y` is the catalogued factual answer used as the prior candidate; the source-erased prompt measures how strongly the checkpoint favors that candidate rather than assuming a strong prior for every item. `X` is a distinct same-type counterfactual, `Z` is a third same-type candidate for the corrupt prompt, and the fourth menu entry is a filler. Candidate order and per-item tags are randomized by the recorded generator stream. The primary fact banks are frozen in `pilot/gen_items.py` (`LOC_FACTS`, `AUTH_FACTS`, and their candidate pools); no model output is used to choose the primary trajectory items. The source-erased prompt keeps the same roster but removes the answer statement, while the neutral diagnostic uses a fictional subject.

For example, one rendered primary pair is:

```
Options: <r3> Austen, <p3> Dante, <y7> Orwell, <q7> Shakespeare.
In this document, Pride and Prejudice was written by Dante.
According to the document, Pride and Prejudice was written by
```

The matched erased version replaces the second line with "The document does not provide this information." and keeps the roster and answer stem unchanged. Three frozen paraphrase templates (A, B, and C) alter the source wording but preserve the candidate menu, answer, and query. For the causal runs, the rendered clean and corrupt prompts are required to have identical token length and exactly one different source-answer token.

3.2 Checkpointed models and transition selection

Table 1 summarizes the checkpoint inventory. OLMo-2 1B is the `allenai/OLMo-2-0425-1B` stage-1 trajectory with 24 checkpoints spanning 21B to 4001B training tokens. OLMo-2 7B is the independent `allenai/OLMo-2-1124-7B` stage-1 trajectory with eight checkpoints spanning 101B to 3896B tokens. The exact released model cards and revisions are recorded for the 1B and 7B checkpoints [21,22]. Models are evaluated without task fine-tuning, in bfloat16 inference mode.

The full OLMo-2 curves use the single reference renderer from the primary generator (`gen_items.generate_pairs` followed by `prior_law.run_one`), with the 24-checkpoint `checkpoints_all.json` manifest for 1B and the eight-checkpoint 7B manifest. They are not averages of the three paraphrase templates. Templates A–C are introduced only in the transition-level behavioral and causal audits after the trajectory checkpoints have been selected; their rows therefore test prompt robustness and intervention behavior rather than define the full curves.

For the two independent-family screens, the scalar gate is computed before any semantic confirmation from the same all-candidate reversion rate on 12 fixed coarse identity checkpoints. For every interior checkpoint, the analyzer selects the minimum reversion among earlier checkpoints and the minimum among later checkpoints, and retains the triplet maximizing the smaller of the two rises. A family passes only when a triplet exists, its observed excursion is at least 0.10, the lower endpoint of its 10,000-draw item-cluster bootstrap interval is greater than 0.05, and both deterministic SHA-256 item-key halves have excursions of at least 0.05. The Pythia coarse axis contains 0, 2, 4, 8, 17, 34, 67, 101, 134, 201, 252, and 300 billion-token labels; the Amber axis contains the 12 public ordinals 0, 8, 16, 32, 64, 96, 128, 176, 224, 272, 320, and 358. Amber labels are ordinal checkpoints, not token counts. Once a family passes, its before/peak/after revisions are frozen and reused for all three semantic axes; no semantic axis selects a new peak. The gate is a screening rule, not a hypothesis test, and checkpoints are not independent statistical replicates.

The trajectory summary defines the excursion of a curve as the largest value of

$$\min\{r(t_p) - r(t_b),\ r(t_p) - r(t_a)\}$$

over an earlier trough t_b, an interior peak t_p, and a later trough t_a. The maximum was selected from the observed curve. For each item-cluster bootstrap draw, the same extrema search is rerun on the resampled curve, so the interval describes the data-selected statistic rather than a confirmatory hypothesis test. The deterministic rerun uses the corrected generator and confirms that the trajectory is not a hash-seed artifact.

The 1B causal event is the late 2496B-to-2999B transition. The global maximum excursion occurs earlier, from 63B through 1909B to 1993B, but the early 1909B-to-1993B drop is not template-general. The late event is therefore the primary intervention transition because all three frozen templates show a positive pairwise Y-over-X error drop. The 7B intervention transition follows the recorded transition-selection rule and is the 3201B-to-3896B source-choice transition; template C is retained in the behavioral audit but is not included in the 7B causal batch because its behavioral change has a confidence interval containing zero.

For the independent-family semantic suite, Pythia-6.9B uses public checkpoint steps 16000, 32000, and 64000 (axis labels 34B, 67B, and 134B) for the pre/peak/post confirmation. Amber-7B uses public LLM360 ordinal checkpoints `ckpt_008`, `ckpt_096`, and `ckpt_272`. Amber ordinals are labels only; no training-token count is imputed. The semantic axes are evaluated at these frozen peaks rather than traced as full semantic trajectories.

Table 1. Checkpoint inventory and item accounting. Amber checkpoint labels are ordinals; no token count is inferred for that release. Behavior and patch counts refer to each semantic condition unless otherwise stated.

Model or suite	Checkpoints used	Checkpoint unit	Item accounting	Purpose
OLMo-2 1B	24	training tokens, 21–4001B	3,840 rows; 121 unique item clusters	deterministic reference trajectory

Model or suite	Checkpoints used	Checkpoint unit	Item accounting	Purpose
OLMo-2 7B	8	training tokens, 101–3896B	1,280 rows; 121 unique item clusters	cross-scale corroboration in a separate release
Pythia-6.9B	12 coarse; 3 semantic confirmation	public training-token labels for coarse identity suite; 34/67/134B for confirmation	120 behavior or 60 patch items per semantic condition	independent family gate and semantic suite
Amber-7B	12 coarse; 3 semantic confirmation	LLM360 ordinal; `ckpt_008/096/272`	120 behavior or 60 patch items per semantic condition	independent family gate and semantic suite

3.3 Source-token activation patching

For every matched causal item, the clean prompt contains source answer `X` and the corrupt prompt changes only that source token to `Z`; the candidate `Y` remains unchanged. We run the clean prompt once with hidden states returned, run the corrupt prompt to obtain its unpatched margin, and identify the unique source-token position after rendering each template. At decoder block l, we replace the residual vector at that position in the corrupt run with the clean vector from the same position and layer. In the Hugging Face outputs used here, `hidden_states[0]` is the embedding output and `hidden_states[k+1]` is the residual stream returned after decoder block `k`. Thus the 1B early-depth summary uses tensor indices 1–12 (blocks 0–11), and the 7B summary uses indices 1–24 (blocks 0–23). This first-75%-of-blocks window is fixed in the patch summary to compare normalized depths across the two OLMo releases; it is a pre-final feature-range convention rather than a transition-specific layer selection. The complete layer curves remain in the intervention readout.

$$q_l = [\ell_l^{\text{patch}}(X) - \ell_l^{\text{patch}}(Y)] - [\ell^{\text{corrupt}}(X) - \ell^{\text{corrupt}}(Y)].$$

Thus q_l is the patch-induced X-versus-Y margin shift relative to the corrupt baseline. It is not bounded by the clean-minus-corrupt margin and can therefore overshoot the clean run. Because `X` and `Y` are single-token answers in this batch, q_l is a final-token logit-difference (one-step log-odds) unit. The primary OLMo transition statistic averages q_l over blocks 0–11 of the 16-layer 1B model or blocks 0–23 of the 32-layer 7B model, then compares the later and earlier checkpoints on paired items.

The nominal causal batches contain 120 rows for 1B and 60 rows for 7B. Conflict batches include repeated draws with the same five-field pair key (`category`, `subject`, `X`, `Z`, `Y`); the canonical paired analyzer indexes these rows by that key, giving 116 unique keys for each 1B conflict comparison and 57 for each 7B comparison. Neutral batches contain 120 and 60 unique keys, respectively. The duplicate-key collapse is fixed by the analysis code and is not based on the observed effect or checkpoint; it explains why the paired estimand can have a smaller n than the nominal file. The source donor is accompanied by a corrupt donor, which copies the corrupt hidden state and is an exact no-op under the implementation. The semantic suite also uses an adjacent clean token as a wrong-position donor and a deterministic norm-matched random residual at the true source position. The latter controls for vector scale but can still produce output changes; it is therefore reported as a diagnostic rather than collapsed into a null. The semantic summary reports the mean of each item's maximum layer shift across all layers; this item-wise peak-layer statistic is descriptive and is not directly interchangeable with the fixed early-depth mean used for the OLMo transition comparison. The current artifact does not include a mismatched natural clean donor or a multi-draw random-donor distribution, so the controls identify positional sensitivity more strongly than source-content specificity.

3.4 Semantic transfer axes

The semantic suite keeps the source-versus-corrupt construction fixed while changing the semantic role of the answer.

- **Type/category:** the passage assigns a counterfactual type to a known entity; item-key halves are assessed separately in the semantic-half check.
- **Relation/slot:** the passage specifies a relation such as headquarters, inventor, birthplace, currency, or language, with relation families and subject identities varied across the item bank.
- **Naturalistic context:** a two-to-four sentence profile or mini-report contains the evidence sentence, aliases, connectives, and distractor details; the query asks for a named location or author slot.

Each axis has three templates, a frozen 50/50 partition of the semantic item keys, and tokenizer-aware candidate construction. The runner stores each key as `axis:category:subject:Y`; for conflict rows this identifies the underlying subject/catalogued-answer fact, while for neutral rows it identifies the frozen fictional-subject/candidate tuple. `analyze_semantic.py` averages repeated rows sharing this key before bootstrap resampling. The split is a deterministic hash of the same key, not a complete category holdout: both prespecified halves are used as robustness checks and in the gate, rather than serving as independent test sets. All answers `X`, `Y`, and `Z` are single first-answer tokens and the clean/corrupt pair differs at exactly one source token. The confirmation suite uses 120 behavior items and 60 patch items per axis, template, and item mode. Conflict and neutral modes are both retained. The analysis specification fixes the two families, three axes, three templates, item split, and pre/peak/post checkpoints before confirmation; the item bank is frozen after tokenizer and schema preflight. No model-based prior-strength threshold or outcome-based item exclusion is applied. Each family-axis input contains the full prespecified matrix, and all inputs in the pooled artifact were retained rather than removed based on effect direction. The semantic preflight is a tokenizer/schema smoke check, not a separate model-based capability filter. `run_semantic_preflight.sh` evaluates 16 items per axis and mode at one frozen revision using template A for the smoke score; the shared item builder validates one-token `X`, `Y`, and `Z` answers and the single clean/corrupt token difference across templates A–C. It rejects malformed or tokenizer-incompatible items but does not apply a prior-strength threshold and does not discard rows according to source gain, patch effect, or any other observed outcome. Accordingly, the confirmation gate has no hidden capability or prior-strength cutoff: it requires complete coverage/schema, at least two of three peak conflict templates with a positive lower 95% confidence bound for source gain, both deterministic item-key halves positive for at least two templates, at least two templates with a positive lower bound for the source-position patch effect, and an exact corrupt-donor no-op within `1e-9`. Because these quality gates and the pooled summaries use the same confirmation files, the pooled values are conditional descriptive summaries, not independent confirmatory tests.

The semantic item builders use frozen fact banks: 19 type facts across three entity categories, 28 relation facts across five relation categories, and the 28 location/authorship facts inherited by the naturalistic axis. In conflict items, the fact-bank answer supplies `Y`; distinct tokenizer-valid pool entries are sampled for `X`, `Z`, and the filler. Neutral items use a fictional subject and sample all four candidates from the valid pool. Candidate order is randomized, and the preflight freezes the bank after tokenizer/schema checks; no model-based prior-strength threshold is applied and no confirmation row is screened by its observed effect. The frozen semantic banks and builder are recorded in `semantic_items.py`; the same renderer is used for behavior, source patching, and controls, so each semantic condition preserves the source/corrupt one-token contrast.

The two independent families are Amber-7B (`IFM/Amber`) [18] and Pythia-6.9B (`EleutherAI/pythia-6.9b-deduped`) [4]. The pooled artifact gives equal weight to families and then equal weight to templates within each family. Model checkpoints are not independent statistical subjects. Each family contributes 36 peak control files for the three axes, three templates, two item modes, and two non-source donor modes; separate corrupt-donor files are retained for every axis and mode.

3.5 Statistical analysis and reproducibility

Trajectory uncertainty is obtained by resampling the 121 unique item clusters with replacement for 10,000 draws. A supplementary sensitivity analysis reweights the same rows by the 28 factual `(category, subject, Y)` clusters and repeats the extrema search and patch summaries with 10,000 fact-cluster draws (full results in `CLUSTER-SENSITIVITY.md`). Fixed item-key halves use a SHA-256 split of the same key; they are deterministic robustness partitions, not independent held-out test sets. Primary causal transition intervals use paired item bootstrap over the matched rows and 10,000 draws. The semantic summaries use the machine-readable 4,000-draw bootstrap over the explicit `axis:category:subject:Y` key defined by `analyze_semantic.py`, averaging repeated rendered rows before resampling, and report family/template confidence checks rather than pseudo-replicating checkpoints.

The prospective predictor analysis was frozen as a gate before the final intervention decision; it was not preregistered. It evaluates checkpoint-by-template groups and reports Bonferroni-adjusted p-values for the 36 pre-final layer, logit, attention, and source-routing tests using the full 48-test family; a post-hoc layer-13 result is included in that 48-test family. A failed gate is retained as a negative result and does not trigger a targeted intervention.

The initial item generator used unordered set iteration for randomized roster identifiers. The deterministic repair preserves random-number-generator order, and a cross-process prompt-hash check gives identical rendered prompts. The semantic runs used a pinned container, disjoint job outputs, and a verified checksum inventory. Full hashes, image metadata, machine provenance, and the 723-entry inventory are provided in the supplementary reproducibility note. These provenance checks validate the frozen artifacts; they do not turn data-selected extrema into fixed confirmatory tests.

4. Results

4.1 Explicit-source reversion is non-monotonic within OLMo-2 trajectories

We first asked whether source reversion could increase and later decrease within one uninterrupted pretraining trajectory. Figure 1 (left) and Figure S1 show the deterministic OLMo-2 1B curve across 24 checkpoints. The largest observed excursion is 0.434, with item-cluster bootstrap 95% CI [0.362, 0.523]. The corresponding data-selected extrema are 63B (before), 1909B (peak), and 1993B (after). The deterministic item halves give excursions 0.439 and 0.462, and the full rerun contains 3,840 rows from 121 unique item clusters. The result is therefore not dependent on one half of the item bank or on the corrected generator's process hash seed.

The maximum excursion is not the transition used for the causal headline. The early 1909B-to-1993B pairwise Y-over-X error drop is positive for template A,

+0.466 [0.379, 0.560], but is negative for template B, -0.095 [-0.172, -0.017], and near zero for template C, +0.009 [-0.078, 0.103]. The later 2496B-to-2999B pairwise Y-over-X error drop is positive for all three templates: A, 0.336 [0.250, 0.431]; B, 0.138 [0.060, 0.216]; and C, 0.397 [0.302, 0.491]. We use this later event as the primary within-trajectory transition because it is the template-robust event in the recorded template check.

Table 2. OLMo-2 trajectory excursions. The extrema are the maximum observed excursion from the audited curve; intervals are item-cluster bootstrap summaries and are not confirmatory tests of a data-selected maximum.

Model	Before	Peak	After	Excursion	95% boot-strap CI	Unique items
OLMo-2 1B	63B	1909B	1993B	0.434	[0.362, 0.523]	121
OLMo-2 7B	1901B	3201B	3896B	0.230	[0.158, 0.298]	121

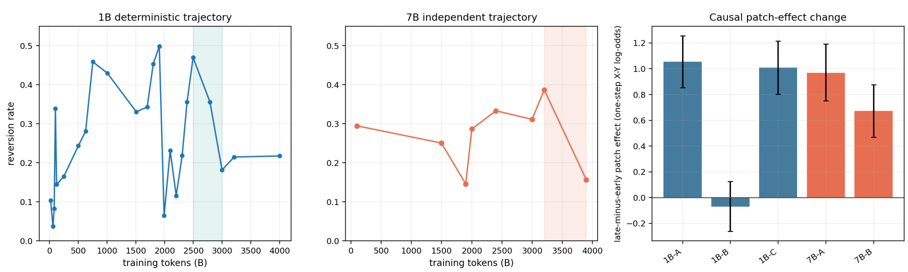

Figure 1. OLMo-2 checkpoint-wise source reversion and selected source-position residual-replacement effects. The left and middle panels show the 1B and separate-release 7B trajectories on a common reversion-rate scale; both full curves use the single reference renderer rather than an average of templates A–C. The right panel shows the change in early-depth source-donor patch effect at the audited transitions, where A–C are transition-level prompt audits. Shaded regions mark the selected late 1B and 7B transitions used for the transition-level audits. Absolute patch effects and paired change intervals are reported in Table 4.

The candidate decomposition shows that the template-robust late event is an exchange between the explicit source and the parametric answer, not merely a change in distractor choices. For 1B, `p_X` rises from 0.390 to 0.762 (change +0.372, 95% CI [0.280, 0.461]) while `p_Y` falls from 0.470 to 0.182 (change -0.288, [-0.382, -0.194]); `p_other` falls from 0.140 to 0.056 (change -0.084, [-0.145, -0.026]). At 7B, `p_X` rises from 0.366 to 0.722 (change +0.355, [0.274, 0.438]) while p_Y falls from 0.387 to 0.157 (change -0.230, [-0.307, -0.156]); p_other falls from 0.247 to 0.121 (change -0.125, [-0.194, -0.061]). Figure S2 shows the full decomposition across all checkpoints; the complete point estimates and item-cluster bootstrap intervals are in choice_decomposition_1b.json and choice_decomposition_7b.json.

The source-erased diagnostics indicate that the competing prior remains present at the selected events. The erased prior-strength summary changes from 6.123 to 6.288 normalized log-likelihood units across the 1B late transition and from 3.324 to 3.302 across the 7B transition. The erased X-minus-Y margin m_e is -6.192 to -6.384 for 1B and -3.436 to -3.285 for 7B, while the source contribution d_m changes from 6.281 to 7.877 and from 3.444 to 4.249, respectively. These continuous diagnostics are descriptive; they are not used to redefine the top-choice decomposition.

Table 3. Candidate-choice decomposition at the selected transitions. Each entry is the mean over the 121 unique item clusters. p_X and p_Y are the proportions whose top-scoring candidate is the explicit source or parametric answer, respectively; p_other is the distractor proportion.

Model	Checkpoint	p_X	p_Y	p_other
OLMo-2 1B	2496B	0.390	0.470	0.140
OLMo-2 1B	2999B	0.762	0.182	0.056
OLMo-2 7B	3201B	0.366	0.387	0.247
OLMo-2 7B	3896B	0.722	0.157	0.121

4.2 The pattern recurs at 7B without an observed shared token boundary

The independent OLMo-2 7B trajectory provides cross-scale corroboration in a separate release with eight checkpoints. Its largest observed excursion is 0.230, with 95% CI [0.158, 0.298], spanning 1901B, 3201B, and 3896B. At the selected 3201B-to-3896B transition, the template audit uses a pairwise Y-over-X error indicator 1{m_c<0}, rather than the all-candidate top-choice reversion used for the trajectory. The raw endpoint rates are computed over the nominal 120 rows; after key-indexing, 116 paired keys are available for the checkpoint contrast. Using those paired keys, the paired error decreases by 0.224 [0.129, 0.319] for template A and by 0.164 [0.086, 0.241] for template B. Template C changes by 0.069 [-0.026, 0.164], so the 7B template audit is partial rather than universal. Raw endpoint rates are retained in the canonical summary; they are not used as the paired estimand, so endpoint subtraction need not equal the paired change. The 116 template-audit and 57 causal-batch counts arise from the fixed five-field duplicate-key aggregation described in Methods, not from outcome-based filtering.

The relative timing is not shared across the two OLMo releases. Progress-aligning the 1B curve to the 7B checkpoints gives a descriptive Spearman

$\rho = -0.048$. Because this comparison uses only eight sparsely sampled, serially dependent checkpoints after alignment, we do not treat the rank correlation as an inferential test of a shared boundary. No shared token-aligned timing was detected across the two trajectories; their sampling does not support estimating a common boundary or a scale law.

4.3 Source-token patching changes the output margin at selected transitions

The behavioral source-choice change could arise from a change elsewhere in the prompt or from an output-layer reweighting that leaves source transmission unchanged. We therefore measured the effect of restoring the clean source-token residual in the corrupt run. Table 4 reports the absolute early-depth mean patch effects at both checkpoints and their paired change from the high-reversion checkpoint to the low-reversion checkpoint.

Table 4. Absolute and change in early-depth source-donor patch effect at the selected transitions. q_{early} and q_{late} are the early-depth mean patch effects at the earlier and later checkpoints; $\Delta q = q_{\text{late}} - q_{\text{early}}$. All effects are single-token X-minus-Y logit-difference (one-step log-odds) units. Bracketed intervals for q_{early} and q_{late} are item-bootstrap 95% confidence intervals; bracketed intervals for Δq are paired item-bootstrap intervals over the common rows. Absolute q means use the available matched rows (116 for conflict and 120 for neutral in the 1B batch; 57 and 60, respectively, in the 7B batch). "Neutral" is a matched prior-minimized diagnostic, not a second conflict endpoint.

Model/mode	Template	Checkpoints	q_{early} [95% CI]	q_{late} [95% CI]	Δq [95% paired CI]
OLMo-2 1B / conflict	A	2496B → 2999B	2.346 [2.048, 2.646]	3.401 [3.105, 3.700]	+1.054 [0.854, 1.255]
OLMo-2 1B / conflict	B	2496B → 2999B	3.034 [2.783, 3.290]	2.964 [2.759, 3.175]	-0.071 [-0.261, 0.125]
OLMo-2 1B / conflict	C	2496B → 2999B	2.264 [2.005, 2.531]	3.273 [2.997, 3.559]	+1.008 [0.802, 1.215]
OLMo-2 1B / neutral	A	2496B → 2999B	3.105 [2.827, 3.385]	4.389 [4.139, 4.643]	+1.284 [1.095, 1.470]
OLMo-2 1B / neutral	B	2496B → 2999B	3.284 [3.018, 3.566]	3.264 [3.057, 3.476]	-0.020 [-0.221, 0.180]
OLMo-2 1B / neutral	C	2496B → 2999B	2.805 [2.563, 3.046]	3.483 [3.254, 3.710]	+0.677 [0.474, 0.888]
OLMo-2 7B / conflict	A	3201B → 3896B	0.849 [0.623, 1.084]	1.817 [1.564, 2.082]	+0.968 [0.752, 1.191]
OLMo-2 7B / conflict	B	3201B → 3896B	1.986 [1.754, 2.219]	2.658 [2.368, 2.954]	+0.672 [0.469, 0.877]
OLMo-2 7B / neutral	A	3201B → 3896B	1.361 [1.117, 1.607]	2.655 [2.305, 3.028]	+1.295 [1.034, 1.569]
OLMo-2 7B / neutral	B	3201B → 3896B	2.243 [1.985, 2.510]	2.742 [2.441, 3.045]	+0.499 [0.299, 0.694]

For OLMo-2 1B, the absolute conflict patch effects for template A are 2.346 [2.048, 2.646] and 3.401 [3.105, 3.700] at the earlier and later checkpoints, respectively; template C has 2.264 [2.005, 2.531] and 3.273 [2.997, 3.559]. Their paired changes are +1.054 [0.854, 1.255] and +1.008 [0.802, 1.215]. Template B changes from 3.034 [2.783, 3.290] to 2.964 [2.759, 3.175] ($\Delta q = -0.071$, [-0.261, 0.125]). The transition is therefore positive for two templates and null for the third. In particular, template B shows a behavioral source-choice return without an increased patch effect, so the local intervention is not a uniform mediator of the observed behavioral change. The 7B result is positive for both tested templates: 0.849 [0.623, 1.084] to 1.817 [1.564, 2.082] ($\Delta q = +0.968$, [0.752, 1.191]) for A and 1.986 [1.754, 2.219] to 2.658 [2.368, 2.954] ($\Delta q = +0.672$, [0.469, 0.877]) for B.

A supplementary fact-cluster sensitivity analysis gives the same qualitative pattern after giving each `(category, subject, Y)` tuple equal weight. The fact-weighted trajectory excursions are 0.482 [0.380, 0.607] for 1B and 0.242 [0.161, 0.321] for 7B, with the same selected extrema. Fact-weighted patch changes remain positive for 1B conflict templates A and C (+1.056 and +1.109) and for both tested 7B conflict templates (+0.997 and +0.656); the 1B conflict template B change remains near zero (-0.155, 95% CI [-0.495, 0.179]). Full fact-cluster intervals for all conflict and neutral groups are provided in `CLUSTER-SENSITIVITY.md`. These are sensitivity estimates from existing rows, not replacements for the item-cluster primary estimands.

The neutral diagnostic helps interpret these changes. At 1B, neutral patch-effect changes are +1.284 [1.095, 1.470] for A, -0.020 [-0.221, 0.180] for B, and +0.677 [0.474, 0.888] for C. At 7B, they are +1.295 [1.034, 1.569] for A and +0.499 [0.299, 0.694] for B. Conflict-minus-neutral differences in differences include zero for 1B templates A and B and 7B templates A and B; the 1B C interaction is +0.331 [0.044, 0.614]. Thus the intervention supports a causal effect of replacing the residual at the tested source position on the measured margin, while the direction is not consistently unique to conflict arbitration.

4.4 Conditional source effects across semantic axes at frozen family peaks

We next asked whether the source-versus-corrupt effect survives a change in semantic role and independent training release. The scalar family gates pass for Pythia-6.9B (excursion 0.212, 95% CI [0.153, 0.306]) and Amber-7B (ordinal excursion 0.182, 95% CI [0.120, 0.252]; half excursions 0.200 and 0.172). The semantic suite then evaluates three axes at the recorded family peaks rather than selecting a new peak separately for each axis.

Table 5 and Figure 2 summarize the equal-family pooled values. Because the semantic quality gate and pooled summaries use the same confirmation files, these are conditional descriptive summaries rather than independent confirmatory tests. Source gain is the clean-minus-corrupt X-versus-Y margin at the peak, in single-token logit-difference units. The source-donor patch effect is the mean, across items, of each item's maximum layer shift from the same corrupt baseline;

it is an item-wise peak-layer descriptive statistic. The pooled source gains are 2.896 for type/category, 3.336 for relation/slot, and 3.742 for naturalistic context. Corresponding source-donor patch effects are 3.251, 3.599, and 3.932. All family/template source-gain confidence checks and all family/template source-donor patch checks are positive. The smallest family/template source-gain lower bounds are 1.314 for Pythia type, 1.358 for Pythia relation, and 2.419 for Pythia naturalistic; the corresponding smallest patch lower bounds are 1.739, 1.619, and 2.636. Amber's minima are 2.198/2.267, 3.061/3.058, and 2.795/2.925 for the same three axes.

Table 5. Equal-family semantic peak summaries. The source-donor patch effect is the mean of item-level maximum layer shifts at the peak. Detailed family-by-template means, confidence intervals, and selected layers are provided in Table S1 (`TABLE-S1-SEMANTIC-DETAILS.md`); the pooled table is not a family-level hypothesis test.

Axis	Measure (X-Y logit diff.)	Amber	Pythia	Equal-family
Type/category	Source gain	3.426	2.366	2.896
Type/category	Patch effect	3.850	2.652	3.251
Relation/slot	Source gain	4.250	2.423	3.336
Relation/slot	Patch effect	4.542	2.656	3.599
Naturalistic	Source gain	3.896	3.588	3.742
Naturalistic	Patch effect	4.208	3.655	3.932

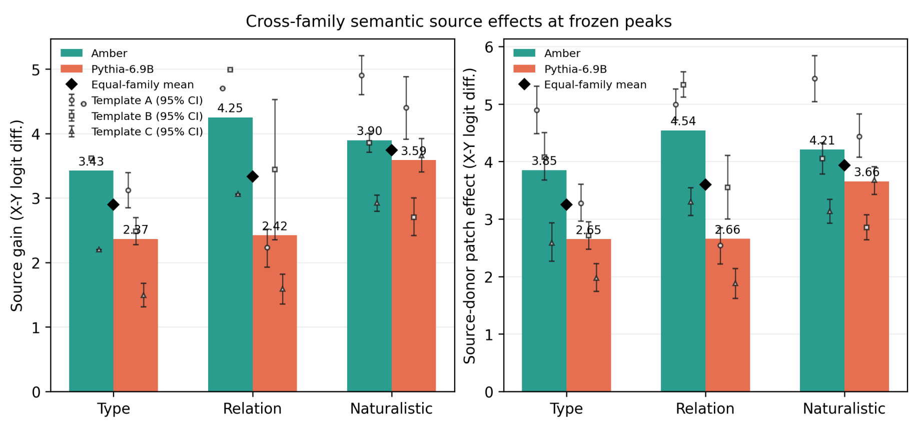

Figure 2. Conditional semantic source gain (left) and source-donor patch effect (right) at frozen peaks for Amber-7B and Pythia-6.9B. Values are in single-token X-minus-Y logit-difference units; bars are equal-template family means, small markers show template estimates with 95% confidence intervals, and diamonds denote equal-family means. The plotted summaries are conditional on the pre-specified semantic quality gate.

The family and template leave-one-out diagnostics preserve the direction. These checks extend the measured source-versus-corrupt effect to the tested semantic roles and document-like prompts at the selected checkpoints. Because each family contributes a small number of correlated checkpoint conditions, the pooled values are descriptive equal-family summaries rather than a claim of independent checkpoint-level replication.

4.5 Controls constrain, but do not erase, alternative explanations

The semantic falsification matrices contain complete coverage, schema, source-position geometry, source/control pairing, and item-multiset checks for both families. Corrupt-donor patches are exact no-ops at every layer. The wrong-position donor is much smaller than the source donor: its patch-effect/ source-donor ratios range from 0.072 to 0.108 across Amber and Pythia axes. This indicates that simply writing an adjacent contextual residual at the source position is not sufficient to reproduce the source effect under the tested geometry.

The norm-matched random-residual control is more informative as a boundary than as a null. Amber ratios range from 0.086 to 0.102, but Pythia ratios are 1.255 for type, 1.253 for relation, and 0.910 for naturalistic context. Random residuals can therefore produce nontrivial output changes in one family. We retain these controls and do not describe the source donor as uniquely identified by the current matrix.

Finally, the prospective predictor gate fails. Across 12 checkpoint-by-template groups, final neutral source gain has a descriptive Spearman $\rho = -0.796$ with the pairwise Y-over-X error. These groups share checkpoints and templates, so the nominal rank-correlation p-value is not used as inferential evidence; none of the 36 pre-final frozen L0–L11 logit-lens, attention, or source-routing tests survives correction. The post-hoc layer-13 analysis also fails its 48-test family. We therefore did not run or claim a targeted-head intervention, early-warning predictor, or steering method.

Table 6. Semantic donor controls and predictor gate. Control values are ratios of the control item-level maximum-layer patch effect to the source-donor patch effect at the same peak. Corrupt donors are exact no-ops and therefore are not assigned a ratio.

Family	Axis	Wrong-position ratio	Random-residual ratio	Interpretation
Amber	Type/category	0.085	0.102	small relative control
Amber	Relation/slot	0.072	0.086	small relative control

Family	Axis	Wrong-position ratio	Random-residual ratio	Interpretation
Amber	Naturalistic	0.078	0.093	small relative control
Pythia-6.9B	Type/category	0.108	1.255	random donor nontrivial
Pythia-6.9B	Relation/slot	0.108	1.253	random donor nontrivial
Pythia-6.9B	Naturalistic	0.078	0.910	random donor nontrivial

The predictor gate comprises 36 pre-final frozen tests at layers 0–11; none survives correction. The descriptive neutral-gain/reversion association is $\rho = -0.796$ across 12 checkpoint-by-template groups; its nominal rank-correlation p-value is not used because the groups share checkpoints and templates.

5. Discussion

5.1 Principal interpretation

Across the tested OLMo-2 trajectories, source-choice behavior is not a monotone function of training progress. The model can move from selecting the explicit source toward a parametric prior and later return toward the source. The late 1B event is robust across all three tested templates, and a related event appears at 7B even though its token location differs. This is a useful empirical property of checkpointed language models: final performance alone can conceal a temporary loss of source-choice behavior.

The activation results add an intervention-based test to this behavioral description. At the selected transitions, replacing the source-token residual in a corrupt run changes the final candidate margin in the predicted direction for the supported templates. The intervention therefore establishes a causal effect of replacing the residual at that source position on the measured output under the stated corruption and patching design. It does not show that the position is the only route by which source information travels, nor does it explain which training events produce the transition.

5.2 Relation to existing accounts of knowledge and context

The source-reversion measure is related to contextual faithfulness [1-3] but is not a factuality score. The conflict answer is intentionally counterfactual, so success means following the supplied source, not agreeing with a presumed world truth.

This distinction lets the experiment isolate source use from whether the passage itself is correct.

The result also complements work showing that hidden states can carry truthfulness or world-state information that is not expressed in the output [8-11]. Those studies motivate the representation/use distinction; our source-token intervention tests the causal effect of replacing a residual at a chosen position and checkpoint. The positive patch effects are thus stronger than a probe correlation for the measured margin, but narrower than a general claim that the model internally "knows" the source.

5.3 A plausible but unestablished account

One interpretation is that pretraining changes the balance between a source signal and a competing parametric prior. The descriptive co-movement of source contribution, prior strength, and reversion in the 1B curve is compatible with that account, and the neutral controls show that the intervention effect is not consistently tied to conflict-specific arbitration. Another possibility is that the transition reflects prompt-format sensitivity or a family-specific change in residual geometry. The current data do not distinguish these hypotheses. The failed prospective predictor is evidence against treating any one measured feature as a reliable gate.

5.4 Scope and implications

The conditional semantic suite shows that the source effect is measurable for category, relation, and short naturalistic context questions in two independent releases at frozen peaks. This makes the result broader than a single identity wording, while keeping the inference auditable: family and template weights are explicit, both prespecified deterministic item-key halves are retained in the gate, and checkpoints are not counted as independent subjects. For model analysis, the practical implication is that source-choice evaluations should include intermediate checkpoints and should separate behavioral source gain from causal internal transmission. For interpretability, the result supports reporting local interventions with corrupt, position, and random-vector controls rather than treating a positive patch curve as a complete circuit explanation.

6. Limitations and boundary conditions

The study has five principal boundaries. First, full semantic trajectories are not available: the three semantic axes are evaluated at frozen peaks selected by the scalar family gate. Second, the model-family evidence consists of OLMo-2, Pythia-6.9B, and Amber-7B, with only two independent families in the semantic pool; a third family and denser cross-size sampling would test generality more strongly. Third, prompts, candidate construction, and the source-token location are controlled but not universal. The early 1B event and 7B template C demonstrate that expression can be prompt-dependent. Fourth, the Pythia random-residual controls can be large, so the present experiments do not establish strict source-content specificity or a unique causal mediator. Fifth, the canonical trajectory and semantic runs do not include the pilot agree condition; capability is therefore anchored by the observed candidate decomposition and matched source-erased/neutral diagnostics rather than by an agree-condition ceiling.

The checkpoint extrema and late-transition choice are evidence-audited rather than preregistered discovery claims. The deterministic rerun, fixed halves, family gates, and leave-one-out checks reduce implementation and selection risk, but they do not replace an independently frozen replication of every future analysis. Finally, the predictor/steering gate failed; no reliable early warning or targeted intervention is reported. These boundaries define the intended follow-up work: dense independent trajectories, intervention designs that separate source content from residual geometry, and circuit-level analyses only after a stable predictor or mediator has been identified.

7. Conclusion

Checkpoint-wise source-reversion analysis identifies a reproducible low–high–low source/prior choice trajectory in the tested OLMo-2 runs. At selected transitions, source-donor residual replacement shifts the answer margin in the predicted direction, and conditional frozen semantic analyses extend the measured source-versus-corrupt effect across three axes in Amber-7B and Pythia-6.9B. Together, the results define a bounded empirical training dynamic and a causal effect of replacing the residual at the tested source position on the measured margin. They also show why donor and position controls are necessary before interpreting a patch as source-content-specific mediation.

8. Reproducibility and availability

The public review release at `https://github.com/gbanyan/source-reversion-training-dynamics` (tag `v0.1.3-review`) contains the analysis scripts, checkpoint manifests, canonical summary JSON, selected raw JSON artifacts, and generated figures. The authoritative semantic pooled summary is `results/canonical/cross_family_semantic_pool.json`; the complete semantic raw matrix is not included in this release. Model weights are obtained from the public OLMo, Pythia, and Amber releases. The release checksum file records the identity of the packaged artifacts.

Supplementary Figures S1–S4 and Tables S1–S2 are supplied separately in `SUPPLEMENTARY.pdf` (source `SUPPLEMENTARY.md`). Their machine-readable values and regeneration notes remain available in `TABLE-S1-SEMANTIC-DETAILS.md`, `CLUSTER-SENSITIVITY.md`, and `cluster_sensitivity.json`.

The main submission figures are paper_results_overview.png and semantic_pool_overview.png. Scientific captions, allowed inferences, and canonical source mappings are collected in `FIGURE-TABLE-CAPTIONS.md`.

The machine-readable semantic detail and fact-cluster sensitivity reports are `TABLE-S1-SEMANTIC-DETAILS.md`, `CLUSTER-SENSITIVITY.md`, and `../results/canonical/cluster_sensitivity.json`.

Declarations

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Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author contributions

Jing-Rung Huang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. Wen-Hsiang Lu: Supervision, Writing – review & editing.

Ethics statement

The study uses publicly released language-model checkpoints and generated text prompts; it does not collect human or animal subject data.

Data and code availability

Analysis code, frozen configuration files, canonical summaries, selected raw JSON artifacts, and figures are available at `https://github.com/gbanyan/source-reversion-training-dyna` (tag `v0.1.3-review`). The repository does not redistribute upstream model weights or third-party corpora. The complete semantic raw matrix is not included in this release; the validated pooled summary and derived tables are public, and access to any restricted raw material follows the applicable data terms.

Generative-AI declaration

During the preparation of this work the authors used generative AI tools (OpenAI's GPT/Codex and Anthropic's Claude) to draft and edit the manuscript text and to check the internal consistency of factual and numerical claims against the underlying experimental results. The authors conceived the study, designed and conducted all experiments, generated and verified all data and results, and made all analytical and interpretive decisions. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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